

VPF-C1T1R1 dvpr - Virtual Battery Factory deliverable

Design Verification Plan and Report (DVP&R) - c1_t1_r1
VPF-C1T1R1-DVPR-001 I Virtual Battery Factory

I # I Requirement I Protocol I Result I Verdict I Evidence I

I---I---I---I---I---I

I 1 I ED >= 392.61 Wh/kg I 1C discharge + calc-energy contract I 502.70 Wh/kg I PASS I cell/r4_v3_ma
I 2 I 4C fast charge without lithium plating I 4C_charge_45C protocol, 20 A, --plating, SPMe I anode
I 3 I T_max <= 60 C in every scenario I lumped thermal, 45 C ambient (4C); overcharge 298.15 K ambie
I 4 I Overcharge to 4.7 V without thermal runaway I 0.5C charge to cut-off + 0.5 V (4.70 V), then ru
I 5 I (backup) all criteria on alternative I same protocols on R4-V1-cool I ED 499.43; anode min +0.
I 6 I (negative control) cooling ceiling I h=150 on R4-V4-coolmax I anode min -0.0006 V -> plated I

Verification method note: verdicts on rows 1-4 are mechanically recorded by
'bda log-evaluate' in log.jsonl (round 4, candidate R4-V3-margin: verdict=pass,
checked=4). Rows 5-6 are the same protocol applied to backup/negative-control variants.